

VANSHU GALHOTRA

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Education

National Institute of Technology, Tiruchirappalli

Aug 2024 – July 2027

*Master of Computer Applications (MCA) - CGPA: 9.44**

Trichy, Tamil Nadu

Panjab University, Chandigarh

Sep 2021 – May 2024

Bachelor of Computer Applications (BCA) - 85.91%

Sector 14, Chandigarh

Experience

Summer Intern, Goldman Sachs

May 2026 – July 2026

Built an MCP Server for a GenAI Support Assistant

- Designed an **MCP server** integrating **LLMs** with **AI Agent orchestration**, exposing read-only, pre-defined database queries as tools.
- Implemented **token-based authentication** and **role-based, tool-level authorization** to secure tool access.
- Developed a **prompt-to-tool evaluation pipeline** to validate LLM tool-selection accuracy, and deployed the application on **Kubernetes**.

Research Intern, DCSA

May 2025 – Jun 2025

Panjab University, Chandigarh

Chandigarh

- Worked on “**Handwritten Trajectories Identification in Scanned Text**”, focused on recovering stroke order from static handwritten images.
- Implemented **graph-theoretic and heuristic-based methods** to extract plausible pen trajectories from scanned images.
- Trained a **neural network** to recognize the recovered feature vector, achieving a recognition accuracy of **98.90%**, at par with existing literature; work led to preparation of a **research paper**.

Projects

Real-time Fleet Management System | *Next.js, NestJS, PostgreSQL, AWS, Nginx*

October 2025 - January 2026

github.com/vanshugalhotra/vehicle-docs-dashboard

- Designed and developed a production-grade real-time fleet management system for a private logistics company to manage vehicle records, document tracking, and operational workflows.
- Built with **Next.js** for the frontend and **NestJS** for the backend, backed by a **PostgreSQL** database.
- Deployed on **AWS** with **Nginx** as a reverse proxy and implemented automated **S3** backups for data reliability; system is currently live and actively used in daily operations.

BackTrack – Reverse Engineering Event (INFOTREK’25) | *Next.js, NestJS, CI/CD, AWS* **May 2025 - June 2025**

github.com/vanshugalhotra/backtrack

- Developed a web platform for INFOTREK’25, an event organised by ACM, to manage and host a reverse-engineering competition.
- Built using **Next.js** for the frontend and **NestJS** for the backend.
- Incorporated **CI/CD pipelines** for automated deployment and a smooth development workflow.

Python Chess Engine | *Python, OOP, Alpha-beta Pruning*

September 2024 - December 2024

pypi.org/project/pychess-engine

- Developed **pychess-engine**, a chess engine library (available on *pypi.org*) written in Python.
- Evaluates chess positions, calculates best moves, and supports integration with various interfaces.
- Follows **OOP principles** for modular design, uses **alpha-beta pruning** for efficient move evaluation, and applies **multi-threading** to improve performance.

Technical Skills

Languages: C, C++, Python, JavaScript, TypeScript, Java (Basic)

Web & Backend: React.js, Next.js, Node.js, NestJS, HTML, CSS, Tailwind CSS, MongoDB, PostgreSQL

Cloud & DevOps: AWS, Docker, Kubernetes, Nginx, CI/CD, Vercel

AI Tooling: Machine Learning, Deep Learning, MCP (Model Context Protocol)

Others: Git, GitHub, OOP, Data Structures & Algorithms (DSA)

Leadership / Extracurricular

Chairperson, ACM Student Chapter

October 2025 – Present

National Institute of Technology, Trichy

Trichy, Tamil Nadu

- Responsible for organizing technical events and workshops under the student chapter; contributed to planning INFOTREK and other activities aimed at building interest in tech among students.

EEC Head, VERSION'26

November 2025 – Present

National Institute of Technology, Trichy

Trichy, Tamil Nadu

- Managed a team to plan, develop, and execute events for VERSION'26.

Achievements

NIMCET Entrance Exam : AIR 10

CUET Entrance Exam : Score 236

BCA, DAV College Chandigarh (Panjab University) : 1st Position

Software Development Competition, Panjab University (Inter College) : 1st Prize

Debugging Competition, Panjab University (Inter College) : 2nd Prize

Mathematics Quiz (Intra College) : 2nd Prize